

ZIYUN NIE

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EDUCATION

East China Normal University

M.S. in Psychology (Developmental Psychopathology)

GPA: 3.58/4

Shanghai, China

Sep. 2023 - Jun. 2026

Nankai University

B.S. in Applied Psychology

GPA: 3.67/4

Tianjin, China

Sep. 2019 - Jun. 2023

PUBLICATION & WORKS IN PROGRESS

Song, Y.*, **Nie, Z.**, & Shan, J. (2024). Comprehension of irony in autistic children: The role of theory of mind and executive function. *Autism Research*, 17(1), 109-124. <https://doi.org/10.1002/aur.3051>

Chen, Z., **Nie, Z.**, Zhao, Y., & Song, Y.* (2026). A meta-analytic review of gaze cueing across the autism trait continuum. *Psychonomic Bulletin & Review*, 33(4). <https://doi.org/10.3758/s13423-026-02882-y>

Nie, Z., Chen, Z.*, & Song, Y.* (2026). Beyond Theory of Mind: Language Processing as a Key Factor for Irony Comprehension in Autistic Children (Under Review)

Kong, L.#, **Nie, Z.#**, Pan, X., & Wang, C.* (2026). Multi-Task Paradigms in Eye-Tracking: Enhancing Diagnostic Social Attention Pattern Detection in Autism Spectrum Disorder via Machine Learning (Under Review; Nie and Kong contributed equally to this work.)

CONFERENCES

Nie, Z., Shan, J., & Song, Y. Comprehension of irony in autistic children: The role of theory of mind and executive function (poster). *2024 Student Academic Annual Conference of the School of Psychological and Cognitive Sciences at East China Normal University, Shanghai, China.*

Nie, Z., Cheng, Q., & Song, Y. Adaptive Learning Challenges in Autism: Reduced Evidence Accumulation and Prior Belief Updating in Probabilistic Reversal Reward Learning. *International Society for Autism Research (INSAR) 2026, Prague, Czech Republic.* (Top-rated abstract)

Nie, Z., Kong, L., Pan, X., & Wang, C. Multi-Task Paradigms in Eye-Tracking: Enhancing Diagnostic Social Attention Pattern Detection in Autism Spectrum Disorder via Machine Learning. *International Society for Autism Research (INSAR) 2026, Prague, Czech Republic.*

RESEARCH APPOINTMENTS

The Hong Kong University of Science and Technology

Research Assistant

Supervisors: Prof. CHENG Chen and Prof. WANG Jingyi

Hong Kong

Jul. 2026 – Present

RESEARCH EXPERIENCE

Understanding irony in autistic children: Theory of mind and executive function Jan. 2023 - Mar. 2024

Co-Leader, East China Normal University

Shanghai, China

- **Led the manuscript writing and revision process**; coordinated feedback among co-authors and successfully published the study in *Autism Research*

- **Designed and conducted data analysis** to examine the relationships among autistic traits, theory of mind, executive function, and irony comprehension (including ironic praise and ironic criticism) using ANOVA, ROC analysis, correlation, regression analysis, and mediation analysis
- Applied **R and Python** for data cleaning, visualization (e.g., ROC curves), and statistical modeling (e.g., regression, mediation analysis)
- Wrote and revised **introduction, method, and result** parts of manuscript, and refining narrative clarity throughout the manuscript

Understanding irony in autistic children: From perception to intervention Oct. 2024 - Present
Leader, East China Normal University Shanghai, China

- Independently designed and managed follow-up experiments to explore cognitive process of irony comprehension in autistic children
- Designed and implemented an **fNIRS hyperscanning** study with adult-child dyads to investigate social-cognitive synchronization of irony processing
- Designed and implemented an **eye-tracking experiment** to examine how social attention affects irony comprehension in autistic children
- Designed a **randomized controlled trial** using theory of mind training to evaluate the intervention effects on irony and metaphor understanding

Enhancing Diagnostic Social Attention Pattern Detection in Autism Spectrum Disorder via Machine Learning and Eye-Tracking Aug. 2025 - Present
Leader, East China Normal University and Nankai University China

- Independently conceptualized, designed, and conducted all data analyses for the study (**N = 148; 75 autistic, 73 TD children**)
- Applied **supervised learning algorithms** with cross-validation and performance tuning for autism classification
- Performed **model interpretability analyses** (feature importance), identifying fixation-based metrics as key features and extending models to predict **symptom severity**
- Conducted **structural analyses (EFA within tasks, CFA/ESEM across tasks)** and found no common latent dimension across different social attention tasks

Adaptive Learning Challenges in Autism: Reduced Evidence Accumulation and Prior Belief Updating in Probabilistic Reversal Reward Learning Feb. 2025 - Present
Leader, East China Normal University China

- Designed and implemented **computational modeling and data analysis** for a probabilistic reversal reward learning (PRRL) study comparing 35 autistic and 41 typically developing children
- Compared **Rescorla–Wagner reinforcement learning and EZ-drift diffusion models (DDM)** to characterize trial-by-trial learning and decision dynamics; adopted the EZ–DDM for final analyses
- Interpreted modeling outcomes to identify **reduced evidence accumulation and prior belief updating** in autistic children, providing insights into adaptive learning mechanisms

Threat-sensitive processing of facial emotions in people with high-AQ Mar. 2024 - Sep. 2025
Leader, East China Normal University Shanghai, China

- Conducted data analysis using **random forest classifiers** and **SHAP** (SHapley Additive exPlanations) to identify how interpretation biases toward specific emotions contribute to individual differences in autistic traits
- Performed **Bayesian regression modeling** using **python** (PyMC) to estimate the process of detecting ambiguous emotion expressions, finding that people with higher autistic traits accumulated sensory evidence faster under ambiguous angry face

- Applied **R and Python** for data cleaning, model implementation, and result visualization (e.g., ROC curves, posterior plots)

Social attention and autistic trait: A meta-analytic study

Nov. 2024 - Apr. 2025

Core member, East China Normal University

Shanghai, China

- **Screened 3,500 research articles with co-authors**, narrowing down to 300 relevant studies based on predefined inclusion criteria
- **Extract variables and information** such as sample characteristics, task paradigms, and outcome measures based on coding frameworks
- Utilized **R (incl. metafor)** for **meta-analytic visualization**, including construction of forest plots and funnel plots to assess effect sizes and potential bias

Emotion recognition and attentional bias in high-AQ populations

Oct. 2022 - Jun. 2023

Leader, Nankai University

Tianjin, China

- **Recruited 25 high-AQ and 31 low-AQ participants from a pool of 306 questionnaires**; designed and conducted two experiments to investigate atypical patterns of facial attention in individuals with high autistic traits.
- Used an **EyeLink eye-tracking system** to record and analyze visual attention metrics (e.g., fixation sequences, AOI transitions) under different facial emotion conditions
- Applied **R (incl. lme4)** for data preprocessing, linear mixed-effects modeling, and visualization across both behavioral and eye-tracking datasets
- Identified **atypical social attention patterns** in high-AQ individuals **using linear mixed models**; under disgust conditions, they fixated significantly fewer interest areas before first looking at the eyes

Evidence-based design of social story picture books for autistic children

Mar. 2021 - Mar. 2022

Core member, Nankai University

Tianjin, China

- **Interviewed 5 special education professionals** to gather insights into the practical needs of autistic children in social skill development
- Developed evidence-based social story narratives and **3 picture book prototypes** targeting key social scenarios
- **Designed an eye-tracking experiment** to assess the cognitive and attentional impact of the intervention materials

LANGUAGE AND SKILLS

Languages: Chinese (native), English (TOEFL: 6.0; R 6.0, L 6.0, S 5.0, W 6.0)

Technical Skills: Python, R language, jsPsych, matlab, E-Prime, SPSS

Techniques: Eye-tracking (EyeLink), fNIRS (NIRx)

TEACHING EXPERIENCES AND VOLUNTEER

- Teaching Assistant, *Developmental Psychology*, East China Normal University (Undergraduate Course, Fall 2024)
- Teaching Assistant, *Mental Health and Services*, Nankai University (Practice Course, Fall 2021); awarded Excellent Undergraduate TA
- Volunteer, Gulian Mango Children's Rehabilitation and Therapy Center (July-August, 2021), China Supported sensory-integration and social-interaction interventions for children with neurodevelopmental disorders.

HONORS & AWARDS

- Outstanding Graduate of East China Normal University, 2026
- First-Class Scholarship for Outstanding Academic Performance (10/180), 2025
- Academic Excellence Scholarship, East China Normal University, 2023–2024 & 2024–2025
- Academic Excellence Scholarship, Nankai University, 2020–2021 & 2022–2023 (awarded to top 10% of cohort)
- China Accreditation Test for Translators and Interpreters (CATTI) Level II (English), Translators Association of China, 2022